

SPACECRAFT-TO-SPACECRAFT ABSOLUTE AND RELATIVE TRACKING USING HIGHER-ORDER NONLINEAR FILTERING

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This work explores relative-absolute state estimation using only relative measurements between spacecraft, without relying on ground-based tracking or known trajectories. The goal is to assess which filtering strategies succeed or fail when estimating both absolute and relative states in the nonlinear cislunar environment governed by the Circular Restricted Three-Body Problem (CR3BP). A second-order Extended Kalman Filter (SOEKF) is implemented by truncating the Higher-Order Nonlinear Extended Kalman Filter (HNEKF) after second-order terms. Two variants are tested: one with second-order dynamics only, and one with second-order terms in both dynamics and measurements. These filters are compared to the standard EKF and Unscented Kalman Filter (UKF) using optical- and radar-based relative measurements. Results show that second-order measurement modeling is essential for accuracy in strongly nonlinear regimes, and demonstrate that higher-order filtering enables stable and accurate spacecraft-to-spacecraft tracking in the CR3BP.

INTRODUCTION

Cislunar space is becoming an increasingly important domain of space activity, especially with NASA's planned Gateway mission, which will host a variety of vehicles in orbit near the Moon. Gateway is expected to operate in a 9:2 resonant near-rectilinear halo orbit (NRHO), providing a staging point for lunar and deep-space missions through docking, resupply, and scientific activity.^{1,2} As the region becomes more populated, the demand for robust spacecraft navigation will increase. The chaotic gravitational environment of the Earth-Moon system, combined with limited visibility from Earth, makes ground-based tracking insufficient for continuous and responsive operations. Autonomous navigation systems that use onboard sensors and relative measurements offer a scalable solution, reducing reliance on Earth-based infrastructure and enabling more flexible mission architectures.³

In this environment, autonomous orbit determination using only relative measurements between spacecraft presents a promising alternative to traditional ground-based tracking. Foundational work by Markley⁴ established the analytical conditions under which absolute observability is possible using relative measurements and identified specific configurations in the two-body problem (2BP) that prevent full state recovery. Building on this, Psiaki⁵ showed that the inclusion of perturbations such as J_2 , along with geometric insights, can significantly expand the observable regime. Hill⁶ introduced a method known as linked, autonomous, interplanetary satellite orbit navigation (Li-AISON Navigation) for autonomous orbit determination using satellite-to-satellite ranging, which

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was later demonstrated in practice during the Capstone mission.⁷ Kaufman⁸ further contributed by proving that full relative observability can be achieved using optical-only measurements. More recently, Greaves and Scheeres^{9–11} demonstrated that maneuvers can be estimated from relative-only measurements, derived formal observability conditions, explored system-level requirements, and showed that an observing spacecraft can estimate both the relative and absolute states of a target using only relative observations.

This work is motivated by the need to enable autonomous spacecraft operations with minimal ground intervention. Eliminating reliance on Earth-based measurements can reduce mission costs and improve robustness. Using relative observations such as range, range-rate, and angular data between nearby spacecraft allows onboard estimation of both the relative and absolute states of the vehicles involved. These capabilities are particularly important for formation flying and navigation in regions like cislunar space, where spacecraft must coordinate around infrastructure like NASA’s Gateway.

We build on prior work by Menges and Scheeres,^{12,13} as well as recent applications of the Higher-Order Nonlinear Extended Kalman Filter (HNEKF),¹⁴ for estimation in cislunar space specifically involving NRHO trajectories.^{15,16} Previous studies primarily used the HNEKF with linear measurement models. Here, we extend this research by examining how the inclusion of higher-order nonlinear terms in measurement models enhances filter performance, particularly in the highly nonlinear dynamical regime of cislunar space. Specifically, we extend the simplified scenario, where the absolute state of a spacecraft on an NRHO was estimated using relative measurements to a known target, to a more challenging scenario in which the absolute states of both spacecraft are unknown and must be jointly estimated using only inter-spacecraft relative observations. This advanced scenario exemplifies spacecraft-to-spacecraft absolute tracking (SSAT), emphasizing robust navigation strategies through optical-based (line-of-sight) and radar-based (range and range-rate) measurements. This work presents one of the first demonstrations of incorporating nonlinear measurement sensitivities into a second-order filter framework for SSAT in the Circular Restricted Three-Body Problem (CR3BP).

The primary objectives of this paper are twofold: first, to explore methodologies for performing relative-absolute estimation using relative measurements, assessing which filtering approaches are most effective and under what conditions they succeed or degrade; second, to develop and apply a higher-order nonlinear filter that incorporates second-order terms in both the dynamics and measurement models, addressing the spacecraft-to-spacecraft absolute and relative tracking problem using relative-only measurements.

The paper begins with background on higher-order dynamics, nonlinear filtering, and the development of higher-order estimation techniques, focusing on the formulation of the HNEKF, which incorporates higher-order terms in both the dynamics and measurement models. The problem setup introduces the CR3BP environment, the modeling of absolute and relative spacecraft dynamics, and different relative measurement models: range, range-rate, and line-of-sight. A representative tracking scenario is then defined, where two spacecraft estimate their absolute and relative states simultaneously using only inter-spacecraft measurements. Monte Carlo simulations are used to compare the performance of the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and two second-order EKF (SOEKF) variants—both derived by truncating the HNEKF after second-order terms—across both optical-based and radar-based measurement models, highlighting the impact of nonlinearities on filter behavior and estimation accuracy.

BACKGROUND

Higher-Order Nonlinear Dynamics

This section reviews the recursive construction and propagation of higher-order sensitivities for nonlinear dynamical systems, following the framework originally developed by Park and Scheeres.¹⁷ These sensitivities are encapsulated in the form of state transition tensors (STTs), which generalize the widely used state transition matrix (STM) to arbitrary orders and provide a means to model the nonlinear evolution of perturbations about a reference trajectory.

Consider a general nonlinear system of ordinary differential equations defined on $\mathbb{R}^n$:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t), \quad (1)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ denotes the system state at time t , n is the dimension of the state vector, and $\mathbf{f}: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is a smooth vector field. The solution of this system, also called the flow map, evolves an initial state $\mathbf{x}_0$ at time t_0 to its propagated value at time t via

$$\mathbf{x}(t) = \phi(t; \mathbf{x}_0, t_0), \quad (2)$$

where ϕ denotes the solution flow of the vector field $\mathbf{f}$. In index notation, this flow is written component-wise as

$$x_i(t) = \phi_i(t; \mathbf{x}_0, t_0), \quad i = 1, 2, \dots, n, \quad (3)$$

where $x_i(t)$ is the i -th component of the state vector and ϕ_i is the corresponding component of the flow.

To describe how small perturbations to the initial state evolve over time, we define the first-order sensitivity:

$$\phi_{(t,t_0)}^{i,j} = \frac{\partial \phi^i(t; \mathbf{x}_0, t_0)}{\partial x_0^j}, \quad (4)$$

where $\phi^{i,j}$ is the (i, j) -th element of the STM, representing the sensitivity of the i -th component of the propagated state with respect to the j -th component of the initial condition. This matrix captures the linearized dynamics about the nominal trajectory and is commonly used in applications such as Kalman filtering, variational analysis, and stability studies.

To extend this concept to capture nonlinear effects, we define the p -th order state transition tensor as:

$$\phi_{(t,t_0)}^{i,\gamma_1 \dots \gamma_p} = \frac{\partial^p \phi^i(t; \mathbf{x}_0, t_0)}{\partial x_0^{\gamma_1} \dots \partial x_0^{\gamma_p}} \quad (5)$$

which encodes the sensitivity of the flow to combinations of multiple partial derivatives with respect to the initial state. These tensors naturally generalize the STM to higher orders and provide a structured way to approximate nonlinear trajectory deviations using Taylor series.

The propagation of each STT is governed by its own recursive differential equation. For example, the differential equations for the first- and second-order STTs are:

$$\begin{aligned} \dot{\phi}^{i,a} &= f^{i,\alpha} \phi^{\alpha,a}, \\ \dot{\phi}^{i,ab} &= f^{i,\alpha\beta} \phi^{\alpha,a} \phi^{\beta,b} + f^{i,\alpha} \phi^{\alpha,ab}, \end{aligned} \quad (6)$$

where the right-hand side contains partial derivatives of the vector field $\mathbf{f}$, defined as:

$$f^{i,a} = \frac{\partial f^i}{\partial x^a}, \quad f^{i,ab} = \frac{\partial^2 f^i}{\partial x^a \partial x^b}. \quad (7)$$

Here, indices $i, a, b, \alpha, \beta \in \{1, \dots, n\}$ denote tensor components and Einstein summation convention. For example:

$$\dot{\phi}^{i,a} = f^{i,\alpha} \phi^{\alpha,a} \equiv \sum_{\alpha=1}^n f^{i,\alpha} \phi^{\alpha,a}. \quad (8)$$

All vector field derivatives in these expressions are evaluated along the nominal trajectory, i.e.,

$$f^{i,a} = \left. \frac{\partial f^i}{\partial x^a} \right|_{\mathbf{x}=\phi(t;\mathbf{x}_0,t_0)}, \quad f^{i,ab} = \left. \frac{\partial^2 f^i}{\partial x^a \partial x^b} \right|_{\mathbf{x}=\phi(t;\mathbf{x}_0,t_0)}. \quad (9)$$

This ensures that the resulting STTs propagate sensitivities with respect to deviations about the actual solution of the nonlinear system.

To be consistent with the initial condition $\phi(t_0; \mathbf{x}_0, t_0) = \mathbf{x}_0$, the STTs are initialized as follows:

$$\phi^{i,j}(t_0, t_0) = \delta^{ij}, \quad (10)$$

$$\phi^{i,jk}(t_0, t_0) = 0, \quad (11)$$

$$\phi^{i,jkl}(t_0, t_0) = 0, \quad \text{for all } p \geq 2, \quad (12)$$

where δ^{ij} is the Kronecker delta function and p is the order of the STTs.

Once the STTs are integrated, they may be used to approximate how a small deviation from the nominal initial condition evolves over time. The deviation of the i -th component at time t , denoted $\delta x^i(t)$, is approximated by the truncated Taylor expansion:

$$\delta x^i(t) \approx \sum_{p=1}^m \frac{1}{p!} \phi_{(t,t_0)}^{i,\gamma_1 \dots \gamma_p} \delta x_0^{\gamma_1} \dots \delta x_0^{\gamma_p}, \quad (13)$$

where δx_0 is the deviation at the initial time and m is the maximum order of the expansion. This formulation provides an accurate way to propagate uncertainty through nonlinear dynamics without resorting to numerical integration of sampled trajectories.

Each term in the series captures a different level of fidelity in modeling the deviation. The first-order term recovers the linearized evolution and is equivalent to the standard STM-based approach. The second-order term introduces curvature effects that improve accuracy in weakly nonlinear regimes. As higher-order terms are included, the expansion more accurately represents the underlying nonlinear dynamics, which is especially important in environments like the Earth-Moon CR3BP where long propagation times and multi-body gravitational interactions lead to strongly nonlinear behavior. In such regimes, higher-order state transition tensors offer substantial improvements in uncertainty modeling, estimation, and control.

Higher-Order Sequential Filtering

Traditional Kalman filtering techniques such as the Linear Kalman Filter (LKF), the EKF, and the UKF provide effective estimation in mildly nonlinear regimes. However, these filters have fundamental limitations. The LKF requires linear system dynamics and measurement models, which

rarely holds in astrodynamics. The EKF linearizes the system about the current estimate but can diverge when nonlinearities are strong or the linearization point is inaccurate.¹⁸ The UKF approximates the mean and covariance propagation using deterministically chosen sigma points and captures some higher-order effects but still has limited ability to fully account for nonlinear coupling between state variables.^{19–22} In practice, the UKF typically achieves no better than second-order accuracy in the propagation of the mean and covariance.

To overcome these limitations, higher-order sequential filters have been developed that explicitly include nonlinear contributions from both the dynamical and measurement models. These filters extend beyond the linear or second-order approximations of the EKF and UKF by using higher-order Taylor expansions of the flow and observation models. Examples include higher-order polynomial filters, higher-order moment filters, and the HNEKF. These approaches enable accurate estimation in scenarios with significant nonlinearities, such as spacecraft motion in the CR3BP.

Higher-Order Numerical Extended Kalman Filter (HNEKF)

One example of a higher-order sequential filtering method is the Higher-Order Numerical Extended Kalman Filter (HNEKF), originally derived by Park and Scheeres.¹⁴ This filter explicitly incorporates higher-order STTs into both the prediction and update steps, allowing it to accurately track uncertainty and state estimates in nonlinear dynamical environments. It builds directly on the orbit uncertainty propagation framework developed in Park and Scheeres¹⁷ and generalizes the classical EKF to use higher-order sensitivities.

Unlike the EKF, which relies only on the first-order STM, the HNEKF uses propagated STTs to evaluate the flow map and measurement functions as high-order expansions. This is especially beneficial in problems such as spacecraft-to-spacecraft tracking in cislunar space, where the dynamics and measurements are highly nonlinear.

The HNEKF follows the classical structure of sequential filters, consisting of a time update (prediction) followed by a measurement update (correction). In the time update, the predicted mean is expressed as:

$$(m_{k+1}^-)^i = \phi^i(t_{k+1}; \mathbf{m}_k^+, t_k) + \delta m_{k+1}^i(\delta \mathbf{x}_k), \quad (14)$$

where the correction term is:

$$\delta m_{k+1}^i(\delta \mathbf{x}_k) = \sum_{p=1}^m \frac{1}{p!} \phi_{(t_{k+1}, t_k)}^{i, \gamma_1 \dots \gamma_p} \mathbb{E} [\delta x_k^{\gamma_1} \dots \delta x_k^{\gamma_p}]. \quad (15)$$

The predicted covariance is given by:

$$\begin{aligned} (P_{k+1}^-)^{ij} &= \sum_{p=1}^m \sum_{q=1}^m \frac{1}{p!q!} \phi_{(t_{k+1}, t_k)}^{i, \gamma_1 \dots \gamma_p} \phi_{(t_{k+1}, t_k)}^{j, \zeta_1 \dots \zeta_q} \mathbb{E} [\delta x_k^{\gamma_1} \dots \delta x_k^{\gamma_p} \delta x_k^{\zeta_1} \dots \delta x_k^{\zeta_q}] \\ &\quad - \delta m_{k+1}^i \delta m_{k+1}^j + Q_k^{ij}, \end{aligned} \quad (16)$$

where Q_k is the process noise covariance.

To evaluate these expressions up to the second-order, closed-form expressions for the state moment expectations are required. Assuming the perturbation $\delta \mathbf{x}_k$ is zero-mean Gaussian with covari-

ance P_k , the central moments, up to the fourth-order, are defined as:

$$\mathbb{E}[\delta x_k^a] = 0, \quad (17)$$

$$\mathbb{E}[\delta x_k^a \delta x_k^b] = P_k^{ab}, \quad (18)$$

$$\mathbb{E}[\delta x_k^a \delta x_k^b \delta x_k^c] = 0, \quad (19)$$

$$\mathbb{E}[\delta x_k^a \delta x_k^b \delta x_k^c \delta x_k^d] = P_k^{ab} P_k^{cd} + P_k^{ac} P_k^{bd} + P_k^{ad} P_k^{bc}. \quad (20)$$

The predicted measurement is expanded similarly:

$$(n_{k+1}^-)^i = h^i(\phi(t_{k+1}; \mathbf{m}_k^+, t_k)) + (\delta n_{k+1}^-)^i, \quad (21)$$

$$(\delta n_{k+1}^-)^i = \sum_{p=1}^m \frac{1}{p!} h_{(t_{k+1}, t_k)}^{i, \gamma_1 \dots \gamma_p} \mathbb{E}[\delta x_k^{\gamma_1} \dots \delta x_k^{\gamma_p}], \quad (22)$$

where h^i is the i -th component of the measurement model and the mapped partial derivatives of the measurement model are propagated, up to the second-order, using:

$$\begin{aligned} h_{(t_{k+1}, t_k)}^{i, a} &= h_{k+1}^{i, \alpha} \phi_{(t_{k+1}, t_k)}^{\alpha, a}, \\ h_{(t_{k+1}, t_k)}^{i, ab} &= h_{k+1}^{i, \alpha} \phi_{(t_{k+1}, t_k)}^{\alpha, ab} + h_{k+1}^{i, \alpha \beta} \phi_{(t_{k+1}, t_k)}^{\alpha, a} \phi_{(t_{k+1}, t_k)}^{\beta, b}, \end{aligned} \quad (23)$$

where

$$h_{k+1}^{i, a} = \frac{\partial h^i}{\partial x^a}, \quad h_{k+1}^{i, ab} = \frac{\partial^2 h^i}{\partial x^a \partial x^b}, \quad (24)$$

and all terms are evaluated at $\mathbf{x} = \phi(t_{k+1}; \mathbf{m}_k^+, t_k)$.

The covariance of the predicted measurement is:

$$\begin{aligned} (P_{k+1}^{\mathbf{zz}})^{ij} &= \left(\sum_{p=1}^m \sum_{q=1}^m \frac{1}{p!q!} h_{(t_{k+1}, t_k)}^{i, \gamma_1 \dots \gamma_p} h_{(t_{k+1}, t_k)}^{j, \zeta_1 \dots \zeta_q} \mathbb{E}[\delta x_k^{\gamma_1} \dots \delta x_k^{\gamma_p} \delta x_k^{\zeta_1} \dots \delta x_k^{\zeta_q}] \right) \\ &\quad - (\delta n_{k+1}^-)^i (\delta n_{k+1}^-)^j + R_{k+1}^{ij}, \end{aligned} \quad (25)$$

$$\begin{aligned} (P_{k+1}^{\mathbf{xz}})^{ij} &= \left(\sum_{p=1}^m \sum_{q=1}^m \frac{1}{p!q!} \phi_{(t_{k+1}, t_k)}^{i, \gamma_1 \dots \gamma_p} h_{(t_{k+1}, t_k)}^{j, \zeta_1 \dots \zeta_q} \mathbb{E}[\delta x_k^{\gamma_1} \dots \delta x_k^{\gamma_p} \delta x_k^{\zeta_1} \dots \delta x_k^{\zeta_q}] \right) \\ &\quad - (\delta m_{k+1}^-)^i (\delta n_{k+1}^-)^j. \end{aligned} \quad (26)$$

In prior HNEKF implementations, the measurement model was often assumed to be linear in the state. Under this assumption, the equations simplify to:

$$\begin{aligned} (n_{k+1}^-)^i &= h^i(\mathbf{m}_{k+1}^-, t_{k+1}), \\ (P_{k+1}^{\mathbf{zz}})^{ij} &= \left(\mathbf{H}_{k+1} \mathbf{P}_{k+1}^- \mathbf{H}_{k+1}^\top + \mathbf{R}_{k+1} \right)^{ij}, \\ (P_{k+1}^{\mathbf{xz}})^{ij} &= \left(\mathbf{P}_{k+1}^- \mathbf{H}_{k+1}^\top \right)^{ij}, \end{aligned} \quad (27)$$

as shown by Park and Scheeres.¹⁴ Regardless of whether the measurement model is linear or nonlinear, the Kalman gain is computed using:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}^{\mathbf{xz}} (\mathbf{P}_{k+1}^{\mathbf{zz}})^{-1}, \quad (28)$$

and the state and covariance are updated with:

$$\begin{aligned} \mathbf{m}_{k+1}^+ &= \mathbf{m}_{k+1}^- + \mathbf{K}_{k+1} (\mathbf{z}_{k+1} - \mathbf{n}_{k+1}^-), \\ \mathbf{P}_{k+1}^+ &= \mathbf{P}_{k+1}^- - \mathbf{K}_{k+1} \mathbf{P}_{k+1}^{\mathbf{zz}} \mathbf{K}_{k+1}^\top, \end{aligned} \quad (29)$$

where $\mathbf{z}_{k+1}$ is the measurement at time t_{k+1} .

This formulation enables the HNEKF to account for nonlinear coupling in both the dynamics and measurements, leading to improved performance over traditional filters in highly nonlinear environments such as cislunar space.

PROBLEM SETUP

Dynamical Model

The Circular Restricted Three-Body Problem

The dynamical model used to propagate each spacecraft's motion in this work is the CR3BP. As discussed in the previous subsection, the agent and target spacecraft evolve independently according to their respective dynamics. In this implementation, the Earth-Moon CR3BP is used to model the motion of both spacecraft under the gravitational effects of the Earth and the Moon.

The CR3BP is a well-established model used to describe the motion of a third body, denoted m_3 , influenced by two primary bodies, m_1 and m_2 , which move in circular orbits about their mutual barycenter. The “restricted” term refers to the assumption that the third body has negligible mass and therefore does not affect the motion of the primaries. The “circular” term refers to the assumption that m_1 and m_2 revolve in circular orbits.^{23,24}

In the Earth-Moon CR3BP, m_1 and m_2 represent Earth and Moon, respectively, while m_3 represents a spacecraft or satellite. The state of the third body is defined as:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{v}(t) \end{bmatrix} \quad (30)$$

where $\mathbf{r}(t)$ and $\mathbf{v}(t)$ are the position and velocity vectors with respect to the Earth-Moon barycenter. A geometric illustration of the CR3BP in the rotating frame is shown in Figure 1.

The pseudopotential used in the CR3BP is defined as:

$$U(x, y, z) = \frac{1}{2} (x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2} \quad (31)$$

where μ is the mass parameter given by:

$$\mu = \frac{m_2}{m_1 + m_2} \quad (32)$$

The quantities r_1 and r_2 represent the distances from the third body to the primaries m_1 and m_2 , respectively, and are defined as:

$$\begin{aligned} r_1(t) &= \sqrt{(x + \mu)^2 + y^2 + z^2} \\ r_2(t) &= \sqrt{(x - 1 + \mu)^2 + y^2 + z^2} \end{aligned} \quad (33)$$

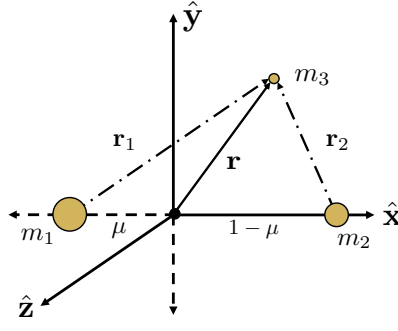


Figure 1. Diagram of the CR3BP in the rotating frame

Using this pseudopotential, the equations of motion in the CR3BP are:

$$\ddot{\mathbf{r}} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 2\dot{y} + \frac{\partial U}{\partial x} \\ -2\dot{x} + \frac{\partial U}{\partial y} \\ \frac{\partial U}{\partial z} \end{bmatrix} \quad (34)$$

These equations define the second-order dynamics of the system and are expressed in state-space form as:

$$\dot{\mathbf{x}}(t) = \frac{d\mathbf{x}}{dt} = \begin{bmatrix} \dot{\mathbf{r}}(t) \\ \dot{\mathbf{v}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{v}(t) \\ \ddot{\mathbf{r}}(t) \end{bmatrix} \quad (35)$$

To support estimation techniques like the LKF and the EKF, the STM $\Phi(t, t_0)$ is used to propagate perturbations in the state from time t_0 to t . It is computed as the Jacobian of the state with respect to the initial state:

$$\Phi(t, t_0) = \frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}(t_0)} \quad (36)$$

This matrix satisfies the first-order differential equation:

$$\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (37)$$

with the initial condition $\Phi(t_0, t_0) = \mathbb{I}_6$. The matrix $A(t)$ is defined as the Jacobian of the dynamics:

$$A(t) = \begin{bmatrix} \mathbf{0}_3 & \mathbb{I}_3 \\ \nabla^2 U & \Omega \end{bmatrix} \quad (38)$$

where $\nabla^2 U$ is the Hessian matrix of the pseudopotential U , and Ω is the Coriolis matrix given by:

$$\Omega = \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (39)$$

which captures the Coriolis accelerations present in the rotating frame of the CR3BP model.

The CR3BP dynamics in Eq. (34) and Eq. (35) are a specific case of the general nonlinear system in Eq. (1), with state $\mathbf{x}(t) \in \mathbb{R}^6$ and vector field $\mathbf{f}(\mathbf{x}, t)$. This identification allows the flow map and its indexed form, $x_i(t) = \phi_i(t; \mathbf{x}_0, t_0)$, to be directly used in constructing and propagating state transition tensors as described in the background section.

Absolute and Relative Spacecraft Dynamics

Consider a system of two spacecraft, an agent spacecraft and a target spacecraft. Both follow ballistic motion in a given dynamical model, such as the Earth-Moon CR3BP. Each spacecraft evolves independently under its own dynamics, meaning there is no dynamical coupling between the two.

The dynamics of the agent spacecraft are defined by a differential equation $\dot{\mathbf{x}}_A = \mathbf{f}(\mathbf{x}_A, t)$, where $\mathbf{x}_A(t)$ is the agent's state at time t . The corresponding flow, propagated from an initial state $\mathbf{x}_{A,0}$ at time t_0 , is given by:

$$\dot{\mathbf{x}}_A = \mathbf{f}(\mathbf{x}_A, t) \rightarrow \mathbf{x}_A(t) = \phi_A(t; \mathbf{x}_{A,0}, t_0). \quad (40)$$

Similarly, the dynamics of the target spacecraft are defined by:

$$\dot{\mathbf{x}}_T = \mathbf{f}(\mathbf{x}_T, t) \rightarrow \mathbf{x}_T(t) = \phi_T(t; \mathbf{x}_{T,0}, t_0). \quad (41)$$

Because the agent and target evolve independently, the relative state between them is defined as:

$$\Delta \mathbf{x}(t) = \mathbf{x}_T(t) - \mathbf{x}_A(t) = \phi_T(t; \mathbf{x}_{T,0}, t_0) - \phi_A(t; \mathbf{x}_{A,0}, t_0). \quad (42)$$

A geometric interpretation of this absolute and relative formulation is shown in Figure 2, where the absolute and relative position and velocity vectors of the agent and target spacecraft are illustrated.

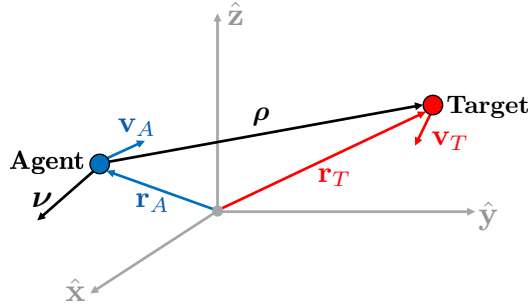


Figure 2. Geometric model of agent and target spacecraft configuration

Using the notation from Figure 2, the absolute state of the agent spacecraft is:

$$\mathbf{x}_A = \begin{bmatrix} \mathbf{r}_A \\ \mathbf{v}_A \end{bmatrix} = [x_A \ y_A \ z_A \ \dot{x}_A \ \dot{y}_A \ \dot{z}_A]^\top, \quad (43)$$

where $\mathbf{r}_A$ and $\mathbf{v}_A$ are the position and velocity vectors of the agent with respect to the system's barycenter, which is the Earth-Moon barycenter in this application.

Similarly, the target spacecraft's absolute state is:

$$\mathbf{x}_T = \begin{bmatrix} \mathbf{r}_T \\ \mathbf{v}_T \end{bmatrix} = [x_T \ y_T \ z_T \ \dot{x}_T \ \dot{y}_T \ \dot{z}_T]^\top, \quad (44)$$

where $\mathbf{r}_T$ and $\mathbf{v}_T$ are the target's position and velocity vectors with respect to the same barycenter.

With both spacecraft's absolute states defined, the relative state, which models the motion of the target with respect to the agent, is expressed as:

$$\Delta \mathbf{x} = \mathbf{x}_T - \mathbf{x}_A = \begin{bmatrix} \boldsymbol{\rho} \\ \boldsymbol{\nu} \end{bmatrix} = [\rho_x \ \rho_y \ \rho_z \ \nu_x \ \nu_y \ \nu_z]^\top, \quad (45)$$

where the relative position and velocity vectors are $\boldsymbol{\rho} = \mathbf{r}_T - \mathbf{r}_A$ and $\boldsymbol{\nu} = \mathbf{v}_T - \mathbf{v}_A$, respectively.

To estimate the absolute and relative motion from relative-only measurements, the absolute states of both spacecraft are combined into a single stacked state:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_T \\ \mathbf{x}_A \end{bmatrix}. \quad (46)$$

Relative Measurement Models

Relative measurements between spacecraft form the basis of autonomous tracking in this work. We consider two separate sensing methods: optical-based observations (line-of-sight vector or angles), and radar-based observations (range and range-rate). In both cases, the agent spacecraft measures the relative motion of a target spacecraft using only inter-spacecraft measurements without external references. These measurements are modeled in the rotating frame of the CR3BP and rely on the relative position and velocity vectors defined in Eq. (45).

Figure 3 illustrates the geometric relationship between the spacecraft and the measured quantities.

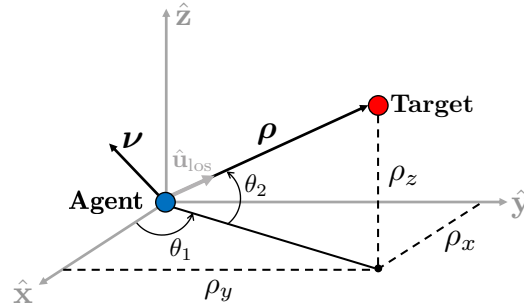


Figure 3. Spacecraft-to-spacecraft relative measurement geometry

Optical-Based Observations

In the optical-only scenario, the agent passively observes the direction of the target spacecraft without access to range information. The primary measurement is the line-of-sight (LOS) unit vector, defined as:

$$\hat{\mathbf{u}}_{\text{los}} = \frac{\boldsymbol{\rho}}{\rho} = \frac{1}{\rho} \begin{bmatrix} x_T - x_A \\ y_T - y_A \\ z_T - z_A \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} \rho_x \\ \rho_y \\ \rho_z \end{bmatrix} \quad (47)$$

This unit vector specifies the direction from the agent to the target and can be constructed from azimuth and elevation angles as follows:

$$\theta_1 = \tan^{-1} \left(\frac{\rho_y}{\rho_x} \right), \quad \theta_2 = \tan^{-1} \left(\frac{\rho_z}{\sqrt{\rho_x^2 + \rho_y^2}} \right) \quad (48)$$

The reverse mapping yields the LOS vector components in terms of these angles:

$$\hat{\mathbf{u}}_{\text{los}} = \begin{bmatrix} \cos \theta_2 \cos \theta_1 \\ \cos \theta_2 \sin \theta_1 \\ \sin \theta_2 \end{bmatrix} \quad (49)$$

The corresponding optical-based measurement function is defined as:

$$\mathbf{h}_{\text{los}}(\mathbf{x}_T, \mathbf{x}_A) = \hat{\mathbf{u}}_{\text{los}} = \frac{1}{\rho} \begin{bmatrix} \rho_x \\ \rho_y \\ \rho_z \end{bmatrix} \quad (50)$$

This model corresponds to an uncooperative scenario in which the target does not transmit signals and is instead tracked through the agent's onboard optical sensors. The agent estimates the relative direction using camera or star-tracker measurements.

Radar-Based Observations

The radar-based measurements consist of the relative range ρ and range-rate $\dot{\rho}$, which are derived directly from the relative position $\boldsymbol{\rho} = \mathbf{r}_T - \mathbf{r}_A$ and relative velocity $\boldsymbol{\nu} = \mathbf{v}_T - \mathbf{v}_A$:

$$\rho = \sqrt{\boldsymbol{\rho}^\top \boldsymbol{\rho}} = \sqrt{\rho_x^2 + \rho_y^2 + \rho_z^2} \quad (51)$$

$$\dot{\rho} = \frac{\boldsymbol{\rho}^\top \boldsymbol{\nu}}{\rho} \quad (52)$$

The radar-based measurement function is then defined as:

$$\mathbf{h}_\rho(\mathbf{x}_T, \mathbf{x}_A) = \begin{bmatrix} \rho \\ \dot{\rho} \end{bmatrix} \quad (53)$$

This model corresponds to cooperative tracking, where both spacecraft transmit and receive active ranging signals.

To support second-order filtering methods, partial derivatives of the measurement functions with respect to the state are required. For instance, considering the radar-based measurement model in Eq. (53), the first-order partial derivatives of the range with respect to the absolute position vectors of the target and agent are:

$$\frac{\partial \rho}{\partial \mathbf{r}_T} = \frac{\boldsymbol{\rho}^\top}{\rho}, \quad \frac{\partial \rho}{\partial \mathbf{r}_A} = -\frac{\boldsymbol{\rho}^\top}{\rho} \quad (54)$$

A second-order partial of the range with respect to the absolute position vector of the target is:

$$\frac{\partial^2 \rho}{\partial \mathbf{r}_T^2} = \frac{1}{\rho} \left(\mathbb{I}_3 - \frac{\boldsymbol{\rho} \boldsymbol{\rho}^\top}{\rho^2} \right) = \frac{1}{\rho} \left(\mathbb{I}_3 - \hat{\mathbf{u}}_{\text{los}} \hat{\mathbf{u}}_{\text{los}}^\top \right) \quad (55)$$

The full expressions for all measurement model derivatives, including optical and range-rate terms, are provided in the Appendix.

SIMULATION CONFIGURATION

The primary goal of the simulation scenario presented here is to evaluate the efficacy of higher-order nonlinear filters, specifically focusing on their application to spacecraft-to-spacecraft absolute and relative tracking using only relative measurements in the nonlinear dynamics environment of cislunar space. To achieve this, numerical simulations were conducted within the Earth–Moon CR3BP framework, modeling both the target spacecraft and the observing agent spacecraft.

In these simulations, the full state vector to be estimated by each filter consists of the absolute translational states of both spacecraft, expressed in the rotating CR3BP frame. This state is defined as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_T \\ \mathbf{x}_A \end{bmatrix} = [\mathbf{r}_T \quad \mathbf{v}_T \quad \mathbf{r}_A \quad \mathbf{v}_A]^\top \in \mathbb{R}^{12}, \quad (56)$$

where $\mathbf{r}_T, \mathbf{v}_T$ represent the position and velocity of the target spacecraft, and $\mathbf{r}_A, \mathbf{v}_A$ represent the position and velocity of the agent spacecraft. Each component is modeled using CR3BP equations of motion, and this 12-dimensional state is estimated simultaneously using relative inter-spacecraft measurements.

The target spacecraft is representative of NASA’s Gateway spacecraft placed in a 9:2 resonant NRHO. The agent spacecraft is located on a quasi-periodic orbit (QPO) characterized by bounded motion relative to the target’s NRHO. This particular QPO exhibits two-dimensional frequency behavior, one frequency of which closely matches the NRHO’s orbital period, ensuring persistent relative phasing between the agent and the target spacecraft.

To amplify nonlinear effects inherent in the dynamics and measurement models, the initial states of the two spacecraft were purposefully placed on opposite sides of the Moon. Specifically, the target spacecraft was initialized near perilune, where the dynamical environment exhibits pronounced nonlinearities due to strong lunar gravitational influences. Conversely, the agent spacecraft was initialized near apolune on its bounded QPO, nearly half a period out of phase relative to the target. This configuration strives to maximize nonlinear measurement and dynamical interactions and is intentionally designed to induce divergence in traditional linear filters, thereby clearly demonstrating the benefits of incorporating higher-order terms in the filter formulation.

The orbits employed in this study are identical to those utilized in previous work by Menges and Scheeres;¹³ however, a significant distinction in this analysis is that the lack of absolute target orbit knowledge aboard the agent spacecraft. This adjustment results in a fully autonomous spacecraft-to-spacecraft tracking scenario in which absolute and relative states of both spacecraft must be estimated simultaneously using solely relative observations. The simulations span eight full orbital revolutions of the Gateway’s 9:2 resonant NRHO.

The orbital configurations of both spacecraft, including their initial states used in the tracking scenario, are depicted in Figure 4.

Two distinct measurement scenarios were explored: an optical-only scenario employing line-of-sight measurements that require no active cooperation from the target, and a radar-only scenario utilizing relative range and range-rate measurements, assuming active cooperation between the spacecraft.

To ensure physical feasibility of the relative measurement models, the simulated scenario was verified to avoid line-of-sight obstructions by the Moon. Specifically, the relative position vectors between the spacecraft were checked to confirm that the line connecting them did not intersect the

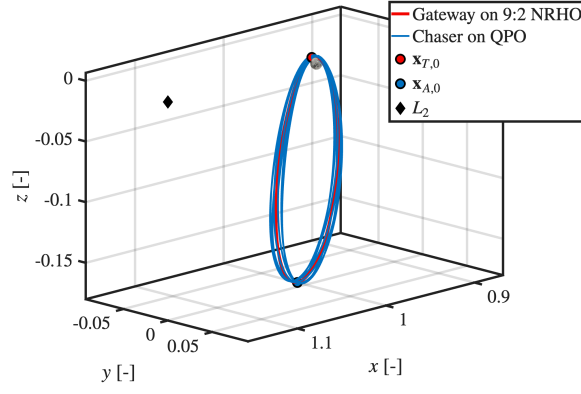


Figure 4. Target spacecraft (NASA’s Gateway) on a 9:2 NRHO and agent spacecraft on a QPO.

Moon’s surface, modeled as a sphere of nondimensional radius $R_m = 1,738 \text{ km}/384,400 \text{ km} \approx 0.00452$, centered at the Moon’s position in the rotating CR3BP frame. No simulated measurements were obstructed under this condition, ensuring all line-of-sight and radar-based observations remained physically valid throughout the simulation.

Initial states for each spacecraft were generated by perturbing the nominal states with Gaussian-distributed deviations:

$$\mathbf{X}_0^* = \mathbf{X}_0 + \begin{bmatrix} \mathbf{w}_{r,\mathbf{x}_T} \\ \mathbf{w}_{v,\mathbf{x}_T} \\ \mathbf{w}_{r,\mathbf{x}_A} \\ \mathbf{w}_{v,\mathbf{x}_A} \end{bmatrix}, \quad (57)$$

where each perturbation vector is sampled from distributions defined as:

$$\begin{aligned} \mathbf{w}_{r,\mathbf{x}_T}, \mathbf{w}_{r,\mathbf{x}_A} &\sim \mathcal{N}(0, \sigma_r) \\ \mathbf{w}_{v,\mathbf{x}_T}, \mathbf{w}_{v,\mathbf{x}_A} &\sim \mathcal{N}(0, \sigma_v), \end{aligned} \quad (58)$$

with uncertainties summarized in Table 1. These values reflect uncertainties previously established for similar spacecraft navigation scenarios in cislunar space.^{25,26}

Table 1. Uncertainty Parameters and Values

Uncertainty Parameters	Values
Initial Position Uncertainty, σ_r	10 km
Initial Velocity Uncertainty, σ_v	0.1 m/s
Range Measurement Uncertainty, σ_ρ	10 m
Range-Rate Measurement Uncertainty, $\sigma_{\dot{\rho}}$	1 mm/s
Line-of-Sight Unit Vector Uncertainty, σ_{los}	1×10^{-5} n.d.
Process Noise (Unmodeled Acceleration), σ_a	1×10^{-14} km/s ²

The initial covariance matrix was defined as:

$$P_0 = \text{blkdiag}(\sigma_r^2 \mathbb{I}_3, \sigma_v^2 \mathbb{I}_3, \sigma_r^2 \mathbb{I}_3, \sigma_v^2 \mathbb{I}_3). \quad (59)$$

Measurement updates were applied daily (every 24 hours), and process noise was modeled as unmodeled accelerations, reflected in the covariance matrix as:

$$Q_k = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^2}{2} \mathbb{I}_3 & \Delta t \mathbb{I}_3 & \frac{\Delta t^2}{2} \mathbb{I}_3 & \Delta t \mathbb{I}_3 \end{bmatrix}^\top \begin{bmatrix} \frac{\Delta t^2}{2} \mathbb{I}_3 & \Delta t \mathbb{I}_3 & \frac{\Delta t^2}{2} \mathbb{I}_3 & \Delta t \mathbb{I}_3 \end{bmatrix}, \quad (60)$$

where Δt represents the interval between measurement updates. The measurement uncertainty matrix R_k was defined as a diagonal matrix with squared uncertainties from Table 1, selected based on the type of measurement model (line-of-sight unit vector for optical, or range and range-rate for radar).

Four filtering methods were analyzed over 200 Monte Carlo simulations: the EKF, UKF, and two second-order EKF (SOEKF) variants derived by truncating the Higher-Order Nonlinear Extended Kalman Filter (HNEKF) after second-order terms. One SOEKF variant assumes a linear measurement model, while the other incorporates second-order terms in both the dynamics and measurement models.

To isolate the impact of measurement model nonlinearity, all filters were tested using identical process noise values chosen to ensure convergence of the full SOEKF. While increasing process noise could artificially stabilize other filters, it degrades estimation accuracy and masks modeling deficiencies. This fixed-noise configuration ensures that performance differences reflect filter formulation rather than noise tuning.

Standard UKF parameters, as specified by Julier and Uhlmann,²² were used and are summarized in Table 2. These parameters were selected to ensure consistent UKF stability under the fixed process noise configuration and to reflect typical values used to handle dynamical nonlinearities. Additional tests confirmed that varying the UKF spread parameter across reasonable ranges did not improve performance; the UKF consistently failed to outperform the full SOEKF under identical noise and measurement settings.

Table 2. UKF Tunable Parameters and Values

Tunable Parameter	Value
Spread Parameter, α	10^{-2}
Distribution Parameter, β	2
Scaling Parameter, κ	0

Although the absolute states of both spacecraft are estimated simultaneously, analyzing the estimation performance is typically more insightful in terms of the target's absolute state and the relative state of the agent with respect to the target. To do this, a linear coordinate transformation is applied to the previously defined combined state vector, $\mathbf{X}$, isolating the target state and the relative state in a target-centered coordinate frame. Specifically, this transformed state vector, $\mathbf{X}'$, is expressed as:

$$\mathbf{X}' = \begin{bmatrix} \mathbf{x}_T \\ \mathbf{x}_A - \mathbf{x}_T \end{bmatrix} = \mathbf{T}\mathbf{X}, \quad (61)$$

where the linear transformation matrix $\mathbf{T}$ is defined as:

$$\mathbf{T} = \begin{bmatrix} \mathbb{I}_6 & \mathbf{0}_6 \\ -\mathbb{I}_6 & \mathbb{I}_6 \end{bmatrix}. \quad (62)$$

Applying this coordinate transformation effectively shifts the analysis to a target-centered reference frame, thereby providing clearer insight into the agent's relative navigation performance. This perspective is particularly useful for interpreting spacecraft-to-spacecraft tracking results, emphasizing the fidelity of relative state estimates and absolute state recovery accuracy of the observed target.

In this transformed coordinate system, the covariance matrix $\mathbf{P}_k$, originally computed in terms of the absolute states of both spacecraft, is similarly transformed via:

$$\mathbf{P}'_k = \mathbf{T}\mathbf{P}_k\mathbf{T}^\top. \quad (63)$$

This provides meaningful covariance bounds centered on the target spacecraft, directly reflecting uncertainty in the target's absolute state and the relative state estimation accuracy, thus aligning the covariance representation with the natural spacecraft-to-spacecraft tracking geometry.

FILTER RESULTS AND DISCUSSION

This section analyzes the filter performance using the transformed target-relative coordinate frame introduced earlier. Results from 200 Monte Carlo simulations are shown for both optical-only and radar-only measurement models, comparing the EKF, UKF, SOEKF with dynamics-only terms, and the full SOEKF. The goal is to assess each filter's ability to handle nonlinearities in both dynamics and measurements under consistent initialization and noise conditions.

Case 1: Optical-Only Measurements

This case considers line-of-sight measurements only, representing a non-cooperative scenario. The goal is to assess filter performance when the measurement model is nearly linear and uncooperative.

Monte Carlo Results for Full SOEKF

Figure 5 shows the estimation error in the target spacecraft's absolute position and velocity magnitudes using the full SOEKF. The error remains within the 3σ bounds throughout the simulation. Although the error does not converge significantly, it remains stable over time. The position error fluctuates between 10 km and 1 km, while the velocity error remains between 10 cm/s and 1 cm/s. These results are expected because line-of-sight measurements do not directly provide absolute information and are relatively insensitive to small improvements in filter formulation.

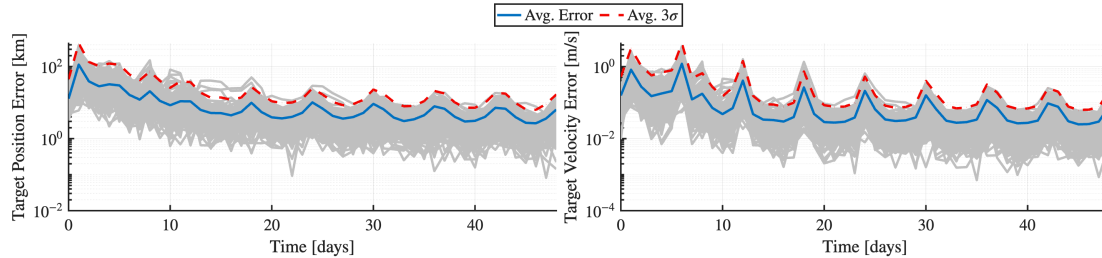


Figure 5. Estimation errors in the target spacecraft's absolute position and velocity magnitudes, shown with 3σ covariance bounds, using the SOEKF with optical-only (uncooperative) relative measurements over 200 Monte Carlo runs.

Figure 6 shows similar results for the relative state between the agent and target. Errors remain within the 3σ bounds and do not show divergence. The relative position error fluctuates between 10 km and 1 km, and the relative velocity error remains between 10 cm/s and 1 cm/s. These results indicate that while high accuracy is difficult to achieve, the filter remains stable in the absence of significant measurement nonlinearity.

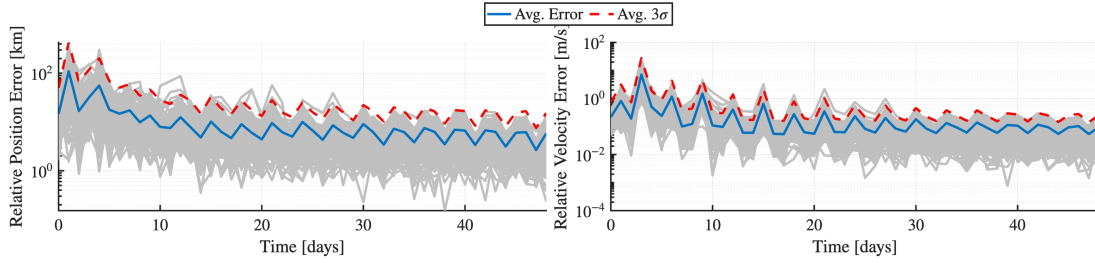


Figure 6. Estimation errors in the relative position and velocity magnitudes between the agent and target spacecraft, shown with 3σ covariance bounds, using the SOEKF with optical-only (uncooperative) relative measurements over 200 Monte Carlo runs.

Comparing with Other Filters

Figure 7 compares all four filters for the target’s absolute state. The EKF diverges early in the simulation. The other three filters—UKF, SOEKF with linear measurement assumption, and full SOEKF—maintain errors within the 3σ bounds. Their performance is nearly identical, suggesting that the measurement model’s linearity is the dominant factor in stability. These filters all handle the nonlinear dynamics well. However, due to the limited information from the line-of-sight measurements, the filters do not achieve high convergence in accuracy. The error remains in the same range: 10 km to 1 km for position and 10 cm/s to 1 cm/s for velocity.

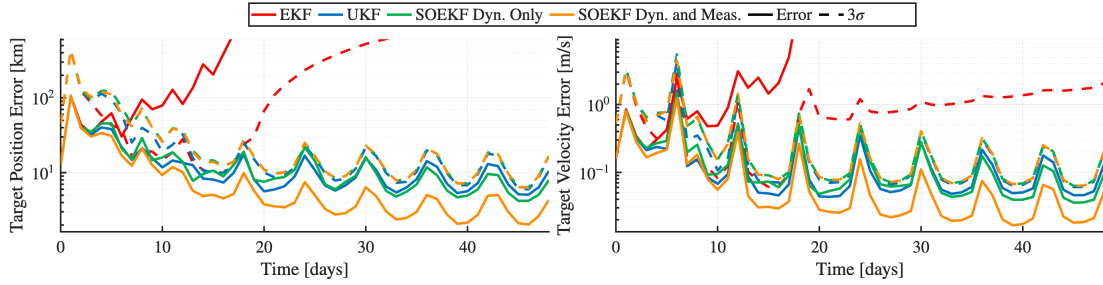


Figure 7. Estimation errors in the target spacecraft’s absolute position and velocity magnitudes, shown with 3σ covariance bounds, comparing the SOEKF to the EKF, UKF, and SOEKF with linear measurement model, using optical-only (uncooperative) relative measurements over 200 Monte Carlo runs.

Figure 8 shows similar results for the relative state. The EKF diverges while the other three filters remain bounded. Their accuracy and behavior are consistent, again highlighting that the measurement nonlinearity is low, and that including second-order dynamics helps prevent divergence. However, convergence is still limited due to the low information content in the measurements.

These results show that line-of-sight measurements alone are not sufficient for achieving high estimation accuracy. However, they offer stable tracking performance when the filter includes higher-order dynamics. The second-order terms in the measurement model do not offer significant improvement due to the near-linearity of the line-of-sight model.

Case 2: Radar-Only Measurements

This case evaluates cooperative measurements, where range and range-rate measurements are used. These measurements introduce stronger nonlinearities but provide more information.

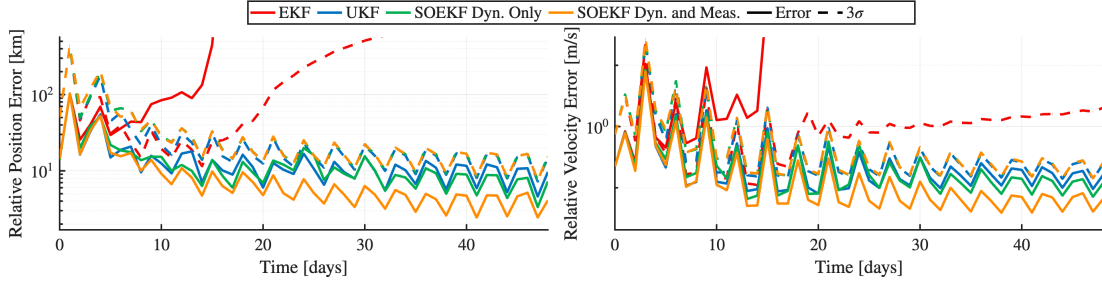


Figure 8. Estimation errors in the relative position and velocity magnitudes between the agent and target spacecraft, shown with 3σ covariance bounds, comparing the SOEKF to the EKF, UKF, and SOEKF with linear measurement model, using optical-only (uncooperative) relative measurements over 200 Monte Carlo runs.

Monte Carlo Results for Full SOEKF

Figure 9 shows that the full SOEKF achieves high accuracy for the target's absolute state. Errors remain within the 3σ bounds, and the average position error decreases from 10 km to 10 m. Velocity errors reduce from 10 cm/s to 0.1 mm/s. These results show that the filter converges successfully due to the higher information content and the second-order terms in both the dynamics and the measurement model, which are necessary to handle the nonlinearities in range and range-rate.

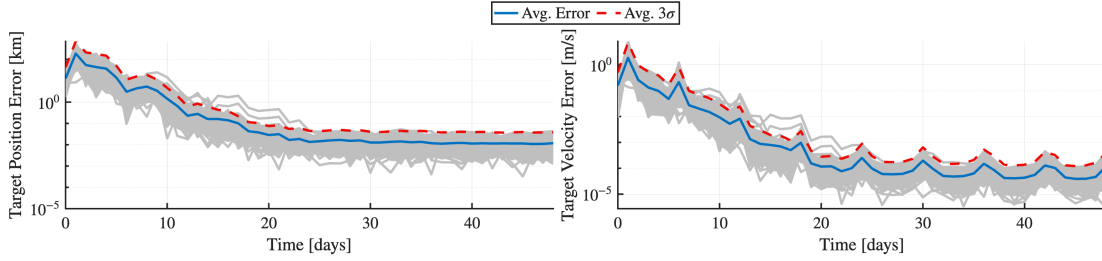


Figure 9. Estimation errors in the target spacecraft's absolute position and velocity magnitudes, shown with 3σ covariance bounds, using the SOEKF with radar-based (cooperative) relative measurements over 200 Monte Carlo runs.

Figure 10 shows similar convergence in the relative state. Position errors drop from 10 km to 10 m, and velocity errors reduce from 10 cm/s to 0.1 mm/s. These improvements are enabled by the radar measurements' higher sensitivity and by modeling second-order effects in both the dynamics and measurement updates.

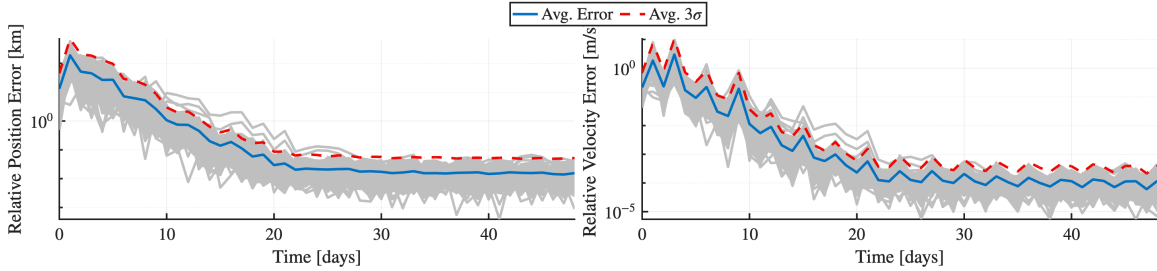


Figure 10. Estimation errors in the relative position and velocity magnitudes between the agent and target spacecraft, shown with 3σ covariance bounds, using the SOEKF with radar-based (cooperative) relative measurements over 200 Monte Carlo runs.

Comparing with Other Filters

Figure 11 shows that the EKF and the SOEKF with linear measurement assumptions diverge quickly. The UKF maintains bounded behavior briefly but diverges later. Only the full SOEKF maintains bounded behavior and achieves convergence, reducing errors by three orders of magnitude. These results show that including higher-order measurement terms is necessary when the measurement model is strongly nonlinear.

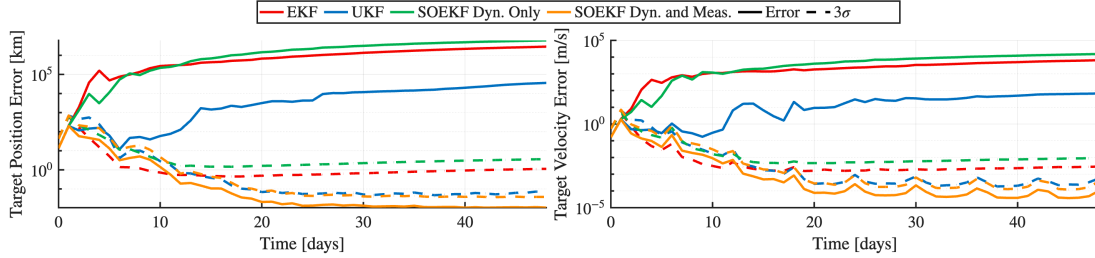


Figure 11. Estimation errors in the target spacecraft's absolute position and velocity magnitudes, shown with 3σ covariance bounds, comparing the SOEKF to the EKF, UKF, and SOEKF with linear measurement model, using radar-only (cooperative) relative measurements over 200 Monte Carlo runs.

Figure 12 shows the same trend in the relative state. Only the full SOEKF remains stable and converges, while the other filters diverge. These results confirm that second-order measurement modeling is required for tracking with nonlinear cooperative measurements like range and range-rate. The UKF diverges in this case, likely due to insufficient approximation accuracy for these nonlinearities.

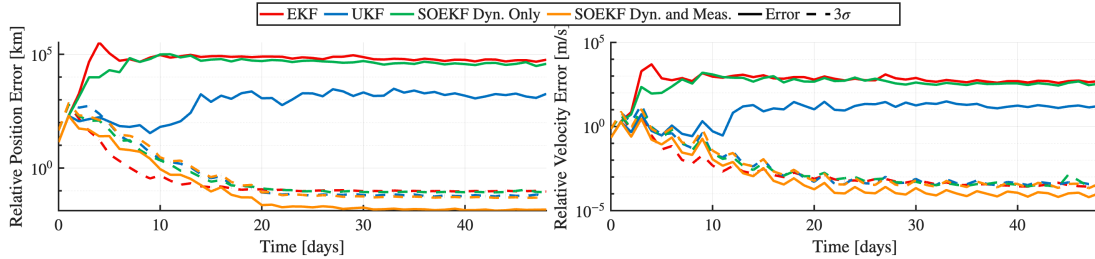


Figure 12. Estimation errors in the relative position and velocity magnitudes between the agent and target spacecraft, shown with 3σ covariance bounds, comparing the SOEKF to the EKF, UKF, and SOEKF with linear measurement model, using radar-only (cooperative) relative measurements over 200 Monte Carlo runs.

Discussion

These simulation results emphasize the impact of filter selection and measurement modeling in spacecraft-to-spacecraft tracking using relative measurements. The study explored the viability of simultaneously estimating absolute and relative states in highly nonlinear dynamical and observational contexts.

For optical-only measurements, the results show that the measurements are nearly linear. While convergence in accuracy is limited, the system remains stable if the nonlinearities in the dynamics are modeled properly. In this case, including second-order measurement terms does not noticeably

impact filter performance. The UKF and the two SOEKF variants perform similarly, and the EKF diverges.

Radar-only measurements provide more information but are highly nonlinear. These results show that the full SOEKF is needed to handle these nonlinearities. Without the second-order measurement terms, the EKF, UKF, and simplified SOEKF all diverge. This confirms that accounting for nonlinear measurement effects is essential in this tracking problem.

The UKF handles dynamical nonlinearities well, as seen in the optical-only case, but fails with nonlinear measurements. Combining both optical and radar data would likely resemble the radar-only case, as radar provides the dominant nonlinear contribution.

All filters were tested using identical process noise values chosen to ensure convergence of the full SOEKF. While increasing the process noise could stabilize the other filters, it would reduce estimation accuracy and obscure deficiencies in the filter formulations. The UKF's spread, distribution, and scaling parameters were selected to maintain stability under optical measurements and were varied across reasonable values without improving radar performance. These results confirm that the UKF's limitations stem from unmodeled measurement nonlinearities rather than poor tuning. The full SOEKF consistently outperformed all other filters under the same conditions, demonstrating robustness to both dynamical and measurement nonlinearities.

Overall, this work demonstrates how higher-order filtering methods, particularly those that include second-order terms in both the dynamics and measurement models, can enable accurate and stable estimation of both absolute and relative spacecraft states in a spacecraft-to-spacecraft tracking scenario using only relative measurements in the CR3BP.

CONCLUSION

This paper investigated spacecraft-to-spacecraft absolute and relative state estimation using only relative measurements in the highly nonlinear dynamics of cislunar space. The main objective was to evaluate various filtering methods, determine the conditions under which they perform well or fail, and highlight the advantages of incorporating higher-order nonlinear terms in both dynamics and measurement models. Two specific measurement scenarios were considered: optical-only (line-of-sight) and radar-only (range and range-rate). The key contribution was the implementation and evaluation of a SOEKF explicitly including second-order nonlinear terms in both the dynamics and measurement models.

The radar-only scenario provided the primary motivation for introducing these higher-order terms, as the range and range-rate measurements introduced significant nonlinearities. In this scenario, the standard EKF, UKF, and the simplified SOEKF (without measurement nonlinearities) quickly diverged. The full SOEKF, by explicitly incorporating second-order measurement partial derivatives, successfully managed these nonlinearities, providing stable convergence and significantly improved accuracy compared to the other filters.

In contrast, the optical-only scenario featured nearly linear measurements. Despite the limited information resulting in lower overall accuracy, the filters demonstrated stable performance. The UKF and both versions of the SOEKF handled the dynamical nonlinearities effectively, showing that proper modeling of dynamics alone is sufficient when measurements are nearly linear. However, due to limited measurement sensitivity, including second-order measurement terms provided little benefit in this scenario.

Additionally, the results provided another robust example of absolute target state estimation using only relative measurements, reinforcing earlier studies. From a theoretical standpoint, explicitly deriving and incorporating second-order measurement partials into the filter architecture was a key outcome. Comparisons between filtering approaches validated these theoretical advancements and clearly illustrated the practical benefits of accurately modeling nonlinear measurement terms.

Overall, this work demonstrated that higher-order filtering methods, particularly those incorporating second-order terms in both dynamics and measurement models, offer substantial advantages for spacecraft-to-spacecraft absolute and relative tracking using relative-only measurements in highly nonlinear regimes such as those encountered in cislunar space.

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APPENDIX: FIRST- AND SECOND-ORDER DERIVATIVES OF RELATIVE MEASUREMENT MODELS

This appendix presents the full set of first- and second-order partial derivatives of the relative measurement models introduced in Eq. (47), Eq. (51), and Eq. (52). These derivatives are necessary for implementing higher-order filtering strategies and are expressed with respect to the absolute target and agent states $\mathbf{x}_T = [\mathbf{r}_T^\top \mathbf{v}_T^\top]^\top$ and $\mathbf{x}_A = [\mathbf{r}_A^\top \mathbf{v}_A^\top]^\top$.

The matrix

$$\mathbf{M} = \mathbb{I}_3 - \hat{\mathbf{u}}_{\text{los}} \hat{\mathbf{u}}_{\text{los}}^\top$$

projects onto the plane orthogonal to the line-of-sight direction. It is used to simplify the first- and second-order derivatives below.

In the second-order derivative expressions, the symbol $\otimes$ denotes the Kronecker product, which constructs a block matrix from the outer product of two vectors or matrices. For example, if $\mathbf{a} \in \mathbb{R}^3$ and $\mathbf{B} \in \mathbb{R}^{3 \times 3}$, then $\mathbf{a}^\top \otimes \mathbf{B} \in \mathbb{R}^{3 \times 9}$ forms a 3-row matrix of 3×3 blocks. Additionally, $\text{vec}(\mathbf{M})$ stacks the columns of $\mathbf{M}$ into a 9×1 vector.

Line-of-Sight Unit Vector $\hat{\mathbf{u}}_{\text{los}}$

First-order partial derivatives:

$$\frac{\partial \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_T} = \frac{1}{\rho} \mathbf{M}, \quad \frac{\partial \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_A} = -\frac{1}{\rho} \mathbf{M}, \quad \frac{\partial \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{v}_T} = \mathbf{0}, \quad \frac{\partial \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{v}_A} = \mathbf{0} \quad (64)$$

Second-order partial derivatives:

$$\begin{aligned}
\frac{\partial^2 \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_T^2} &= -\frac{1}{\rho^2} \left[\hat{\mathbf{u}}_{\text{los}}^\top \otimes \mathbf{M} + \mathbf{M} \otimes \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\text{vec}(\mathbf{M}))^\top \right] \\
\frac{\partial^2 \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_A^2} &= -\frac{1}{\rho^2} \left[\hat{\mathbf{u}}_{\text{los}}^\top \otimes \mathbf{M} + \mathbf{M} \otimes \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\text{vec}(\mathbf{M}))^\top \right] \\
\frac{\partial^2 \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_T \partial \mathbf{r}_A} &= \frac{1}{\rho^2} \left[\hat{\mathbf{u}}_{\text{los}}^\top \otimes \mathbf{M} + \mathbf{M} \otimes \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\text{vec}(\mathbf{M}))^\top \right] = \frac{\partial^2 \hat{\mathbf{u}}_{\text{los}}}{\partial \mathbf{r}_A \partial \mathbf{r}_T}
\end{aligned} \tag{65}$$

All second partial derivatives involving $\mathbf{v}_T$ and $\mathbf{v}_A$ are zero.

Range ρ

First-order partial derivatives:

$$\frac{\partial \rho}{\partial \mathbf{r}_T} = \hat{\mathbf{u}}_{\text{los}}^\top, \quad \frac{\partial \rho}{\partial \mathbf{r}_A} = -\hat{\mathbf{u}}_{\text{los}}^\top, \quad \frac{\partial \rho}{\partial \mathbf{v}_T} = \mathbf{0}, \quad \frac{\partial \rho}{\partial \mathbf{v}_A} = \mathbf{0}. \tag{66}$$

Second-order partial derivatives:

$$\begin{aligned}
\frac{\partial^2 \rho}{\partial \mathbf{r}_T^2} &= \frac{1}{\rho} \mathbf{M}, \quad \frac{\partial^2 \rho}{\partial \mathbf{r}_A^2} = \frac{1}{\rho} \mathbf{M}, \\
\frac{\partial^2 \rho}{\partial \mathbf{r}_T \partial \mathbf{r}_A} &= -\frac{1}{\rho} \mathbf{M} = \frac{\partial^2 \rho}{\partial \mathbf{r}_A \partial \mathbf{r}_T}
\end{aligned} \tag{67}$$

All second partial derivatives involving $\mathbf{v}_T$ and $\mathbf{v}_A$ are zero.

Range-Rate $\dot{\rho}$

First-order partial derivatives:

$$\frac{\partial \dot{\rho}}{\partial \mathbf{r}_T} = \frac{1}{\rho} \boldsymbol{\nu}^\top \mathbf{M}, \quad \frac{\partial \dot{\rho}}{\partial \mathbf{r}_A} = -\frac{1}{\rho} \boldsymbol{\nu}^\top \mathbf{M}, \quad \frac{\partial \dot{\rho}}{\partial \mathbf{v}_T} = \hat{\mathbf{u}}_{\text{los}}^\top, \quad \frac{\partial \dot{\rho}}{\partial \mathbf{v}_A} = -\hat{\mathbf{u}}_{\text{los}}^\top.$$

Second-order partial derivatives:

Position-Position:

$$\frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_T^2} = -\frac{1}{\rho^2} \left[(\mathbf{M} \boldsymbol{\nu}) \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\boldsymbol{\nu}^\top \mathbf{M}) + (\hat{\mathbf{u}}_{\text{los}}^\top \boldsymbol{\nu}) \mathbf{M} \right] \tag{68}$$

$$\frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_A^2} = -\frac{1}{\rho^2} \left[(\mathbf{M} \boldsymbol{\nu}) \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\boldsymbol{\nu}^\top \mathbf{M}) + (\hat{\mathbf{u}}_{\text{los}}^\top \boldsymbol{\nu}) \mathbf{M} \right] \tag{69}$$

$$\frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_T \partial \mathbf{r}_A} = \frac{1}{\rho^2} \left[(\mathbf{M} \boldsymbol{\nu}) \hat{\mathbf{u}}_{\text{los}}^\top + \hat{\mathbf{u}}_{\text{los}} (\boldsymbol{\nu}^\top \mathbf{M}) + (\hat{\mathbf{u}}_{\text{los}}^\top \boldsymbol{\nu}) \mathbf{M} \right] = \frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_A \partial \mathbf{r}_T} \tag{70}$$

Position-Velocity:

$$\begin{aligned}
\frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_T \partial \mathbf{v}_T} &= \frac{1}{\rho} \mathbf{M}, \quad \frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_T \partial \mathbf{v}_A} = -\frac{1}{\rho} \mathbf{M}, \\
\frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_A \partial \mathbf{v}_T} &= -\frac{1}{\rho} \mathbf{M}, \quad \frac{\partial^2 \dot{\rho}}{\partial \mathbf{r}_A \partial \mathbf{v}_A} = \frac{1}{\rho} \mathbf{M}
\end{aligned} \tag{71}$$

Velocity-Position:

$$\begin{aligned}\frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_T \partial \mathbf{r}_T} &= \frac{1}{\rho} \mathbf{M}, & \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_T \partial \mathbf{r}_A} &= -\frac{1}{\rho} \mathbf{M}, \\ \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_A \partial \mathbf{r}_T} &= -\frac{1}{\rho} \mathbf{M}, & \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_A \partial \mathbf{r}_A} &= \frac{1}{\rho} \mathbf{M}\end{aligned}\tag{72}$$

Velocity-Velocity:

$$\frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_T^2} = \mathbf{0}, \quad \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_T \partial \mathbf{v}_A} = \mathbf{0}, \quad \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_A \partial \mathbf{v}_T} = \mathbf{0}, \quad \frac{\partial^2 \dot{\rho}}{\partial \mathbf{v}_A^2} = \mathbf{0}\tag{73}$$